Listing 9.10 Highlighting nested data elements

```
  function hover(hoverD) {
    d3.selectAll("circle").filter(function (d) {return d == hoverD})
             .style("fill", "#94B8FF");
    d3.selectAll("div.datarow").filter(function (d) {return d == hoverD})
             .style("background", "#94B8FF");
            d3.selectAll("rect.bar").filter(function(d) {
                        return d.values.indexOf(hoverD) > -1;
                }).style("fill", "#94B8FF");
   };
```

But we also want to check if the group of tweets represented in each bar contains the data point represented by the element being hovered over (whether a circle or spreadsheet row). We'll use JavaScript's built-in `indexOf` function of an array to test the `rect.bar` elements to see if they contain one of these tweets. We'll also do the same in reverse to highlight the data rows that appear in a level 1 circle of the circle pack or one of the bars in the bar chart. The new, improved functions are shown in the following listing, and the results are shown in figure 9.7.

Listing 9.11 Improved highlighting functions

```
d3.selectAll("div.datarow,circle.pack,rect.bar")
    .on("mouseover", hover)
    .on("mouseout", mouseOut);

function hover(hoverD) {
    var nestArray = hoverD.values || [];
    d3.selectAll("circle.pack")
      .filter(function (d) {return d == hoverD;})
      .style("fill", "#94B8FF");

    d3.selectAll("rect.bar")
    .filter(function (d) {
            return d == hoverD || d.values.indexOf(hoverD) > -1;
    })
    .style("fill", "#94B8FF");

    d3.selectAll("div.datarow")
    .filter(function (d) {
            return d == hoverD || nestArray.indexOf(d) > -1;
    })
    .style("background", "#94B8FF");
};

function mouseOut() {
    d3.selectAll("circle")
    .style("fill", function(d) {return depthScale(d.depth)});

    d3.selectAll("rect").style("fill", "gray").style("stroke-width", 0);
    d3.selectAll("div.datarow").style("background", "white");
};
```

Not every element we mouseover will have a values array, so we need to create an empty array so the indexOf check doesn't fail.

Highlight any rectangle that either shares the same datapoint or has the datapoint in its values array

Highlight any row that shares the same datapoint or is in the values array of the called data